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Electronic component and method of manufacturing the same

Abstract

One object of the present invention is to provide an electronic component less prone to migration of impurity atoms between a conductor and an external electrode. A coil component as an electronic component includes a base body, a conductor, a first external electrode electrically connected to one end portion of the conductor, a second external electrode electrically connected to the other end portion of the conductor, a first film positioned between one end portion of the conductor and the first external electrode, and a second film positioned between the other end portion of the conductor and the second external electrode. The diffusion velocity in the first film and the second film is lower than that in the first external electrode and the second external electrode.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based on and claims the benefit of priority from Japanese Patent Application Serial No. 2019-238965 (filed on Dec. 27, 2019), the contents of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

(2) The present disclosure relates to an electronic component and a method of manufacturing the electronic component.

BACKGROUND

(3) An example of a known electronic component is a coil component such as an inductor. A conventional coil component typically includes a magnetic base body made of a magnetic material, a conductor provided in the magnetic base body and wound around a coil axis, and an external electrode connected to an end portion of the conductor. Such a coil component is mounted on a substrate, for example, through electric connection between the external electrode and the substrate soldered to each other, for use as a component of various electronic devices. An example of the conventional coil component is disclosed in Japanese Patent Application Publication No. 2019-140371.

(4) Between the conductor and the external electrode of the electronic component, migration of impurity atoms contained in the solder or the like may occur due to heat or voltage application. Once the migration occurs, the impurity atoms are alloyed with the material forming the conductor or the external electrode, thereby forming voids in the conductor and/or the external electrode. As a result, the joint strength between the conductor and the external electrode is unfavorably reduced.

SUMMARY

(5) One object of the present invention is to provide an electronic component and a method of manufacturing the electronic component less prone to migration of impurity atoms between the conductor and the external electrode. Other objects of the present invention will be made apparent through the entire description in the specification.

(6) An electronic component according to one embodiment of the present invention comprises: a base body; a conductor provided inside or outside the base body; a first external electrode and a second external electrode both electrically connected to the conductor; and a first film positioned between the first external electrode and one end portion of the conductor electrically connected to the first external electrode, and a second film positioned between the second external electrode and the other end portion of the conductor electrically connected to the second external electrode, wherein a diffusion velocity in the first film and the second film is lower than that in the first external electrode and the second external electrode.

(7) In one embodiment of the present invention, a thickness of the first film and a thickness of the

second film may be 10 nm to 200 nm.

(8) In one embodiment of the present invention, the first film and the second film may be oxide films.

(9) In one embodiment of the present invention, the first film and the second film may be an oxide of a metal material having the same or a lower ionization tendency than a material forming the conductor.

(10) In one embodiment of the present invention, the first film and the second film may be nitride films, carbonized films, or oxynitride films.

(11) In one embodiment of the present invention, the first film and the second film may be amorphous.

(12) In one embodiment of the present invention, indentations in an interface between the first film and the first external electrode and an interface between the second film and the second external electrode may be larger than indentations in surfaces of the first external electrode and the second external electrode.

(13) In one embodiment of the present invention, the conductor may include a portion wound around a coil axis.

(14) An embodiment of the present invention relates to a circuit board comprising any one of the above coil components. One embodiment of the present invention relates to an electronic device comprising the above circuit board.

(15) A method of manufacturing an electronic component according to one embodiment of the present invention is a method of manufacturing an electronic component including a base body, a conductor provided inside or outside the base body, and a first external electrode and a second external electrode both electrically connected to the conductor, the method comprising: a first step of positioning a first end portion and a second end portion of the conductor at a surface of the base body, the first end portion being configured to be electrically connected with the first external electrode, the second end portion being configured to be electrically connected with the second external electrode; a second step of forming a first film and a second film, the first film being electrically connected with the first end portion, the second film being electrically connected with the second end portion; and a third step of forming a first external electrode and a second external electrode, the first external electrode being electrically connected with the first film, the second external electrode being electrically connected with the second film, wherein in the second step, the first film and the second film are formed of a material having a lower diffusion velocity than a material forming the first external electrode and the second external electrode.

Advantageous Effects

(16) The present invention provides an electronic component and a method of manufacturing the electronic component less prone to migration of impurity atoms between the conductor and the external electrode.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view schematically showing a coil component as an electronic component according to one embodiment of the present invention.

(2) FIG. 2 is an enlarged sectional view schematically showing, on an enlarged scale, a sectional surface of a magnetic base body of the coil component shown in FIG. 1.

(3) FIG. 3 is an enlarged sectional view showing, on an enlarged scale, a sectional surface around the joint between one end portion of a conductor and an external electrode in the coil component shown in FIG. 1.

(4) FIG. 4 is a schematic view showing a cross-sectional SEM image of the joint between an end

surface of the conductor and a film in the coil component shown in FIG. 1.

(5) FIG. 5 is a perspective view schematically showing a coil component according to another embodiment of the present invention.

(6) FIG. 6 is a sectional view schematically showing a capacitor as an electronic component according to still another embodiment of the present invention.

DESCRIPTION OF THE EMBODIMENTS

(7) Various embodiments of the present invention will be hereinafter described with reference to the accompanying drawings. The constituents common to more than one drawing are denoted by the same reference signs throughout the drawings. For convenience of explanation, the drawings are not necessarily drawn to scale.

(8) A coil component **1** as an electronic component according to one embodiment of the present invention will be hereinafter outlined with reference to FIG. 1. FIG. 1 is a perspective view schematically showing the coil component **1**. As shown in FIG. 1, the coil component **1** includes a base body **10**, a coil conductor **25** provided in the base body **10**, an external electrode (first external electrode) **21** disposed on a surface of the base body **10**, and an external electrode (second external electrode) **22** disposed on the surface of the base body **10** at a position spaced apart from the external electrode **21**.

(9) In this specification, a “length” direction, a “width” direction, and a “thickness” direction of the coil component **1** correspond to the “L axis” direction, the “W axis” direction, and the “T axis” direction in FIG. 1, respectively, unless otherwise construed from the context. The “thickness” direction is also referred to as the “height” direction.

(10) The coil component **1** is mounted on a circuit board (not shown). The circuit board has two land portions provided thereon. The coil component **1** is mounted on the circuit board by bonding the external electrodes **21**, **22** to the land portions corresponding to the external electrodes **21**, **22**, respectively. The circuit board can be installed in electronic devices such as smartphones, tablets, game consoles, and various others. The circuit board may also be installed in an electric component of an automobile, which is a sort of electronic device.

(11) The coil component **1** may be applied to inductors, transformers, filters, reactors, and various other coil components. The coil component **1** may also be applied to coupled inductors, choke coils, and various other magnetically coupled coil components. Applications of the coil component **1** are not limited to those explicitly described herein.

(12) The base body **10** is made of an insulating material. In one embodiment, the magnetic base body **10** is made mainly of a magnetic material and formed in a rectangular parallelepiped shape. In the coil component **1** according to one embodiment of the invention, the base body **10** has a length (the dimension in the L axis direction) of 1.0 mm to 4.5 mm, a width (the dimension in the W axis direction) of 0.5 mm to 3.2 mm, and a height (the dimension in the T axis direction) of 0.5 mm to 5.0 mm. The dimensions of the base body **10** are not limited to those specified herein. The term “rectangular parallelepiped” or “rectangular parallelepiped shape” used herein is not intended to mean solely “rectangular parallelepiped” in a mathematically strict sense.

(13) The base body **10** has a first principal surface **10a**, a second principal surface **10b**, a first end surface **10c**, a second end surface **10d**, a first side surface **10e**, and a second side surface **10f**. These six surfaces define the outer periphery of the base body **10**. The first principal surface **10a** and the second principal surface **10b** are at the opposite ends in the height direction, the first end surface **10c** and the second end surface **10d** are at the opposite ends in the length direction, and the first side surface **10e** and the second side surface **10f** are at the opposite ends in the width direction.

(14) As shown in FIG. 1, the first principal surface **10a** lies on the top side of the magnetic base body **10**, and therefore, the first principal surface **10a** may be herein referred to as “the top surface.” Similarly, the second principal surface **10b** may be referred to as “the bottom surface.” The coil component **1** is disposed such that the first principal surface **10a** faces the circuit board, and therefore, the first principal surface **10a** may be herein referred to as “the mounting surface.”

The top-bottom direction of the coil component **1** refers to the top-bottom direction in FIG. **1**.

(15) Next, the base body **10** which is magnetic will be further described with reference to FIG. **2**. FIG. **2** is an enlarged sectional view schematically showing, on an enlarged scale, a sectional surface of the base body **10**. As shown in the drawing, the base body **10** contains a plurality of first metal magnetic particles **11**, a plurality of second metal magnetic particles **12**, and a binder **13**. The binder **13** binds together the plurality of first metal magnetic particles **11** and the plurality of second metal magnetic particles **12**. In other words, the base body **10** is formed of the binder **13** and the plurality of first metal magnetic particles **11** and the plurality of second metal magnetic particles **12** bound to each other by the binder **13**.

(16) The plurality of first metal magnetic particle **11** have a larger average particle size than the plurality of second metal magnetic particles **12**. That is, the average particle size of the plurality of first metal magnetic particles **11** (hereinafter referred to as the first average particle size) is different from the average particle size of the plurality of second metal magnetic particles **12** (hereinafter referred to as the second average particle size). For example, the first average particle size is 30 μm , and the second average particle size is 0.1 μm , but these are not limitative. In one embodiment of the present invention, the magnetic base body **10** may further contain a plurality of third metal magnetic particles (not shown) having an average particle size different from the first average particle size and the second average particle size (the average particle size of the third metal magnetic particles is hereinafter referred to as the third average particle size). The third average particle size may be smaller than the first average particle size and larger than the second average particle size, or it may be smaller than the second average particle size. The first metal magnetic particles **11**, the second metal magnetic particles **12**, and the third metal magnetic particles contained in the magnetic base body **10** may be hereinafter collectively referred to as “the metal magnetic particles” when they need not be distinguished from one another.

(17) The first metal magnetic particles **11** and the second metal magnetic particles **12** can be formed of various soft magnetic materials. For example, a main ingredient of the first metal magnetic particles **11** is Fe. Specifically, the first metal magnetic particles **11** are particles of (1) a metal such as Fe or Ni, (2) a crystalline alloy such as an Fe—Si—Cr alloy, an Fe—Si—Al alloy, or an Fe—Ni alloy, (3) an amorphous alloy such as an Fe—Si—Cr—B—C alloy or an Fe—Si—Cr—B alloy, or (4) a mixture thereof. The composition of the metal magnetic particles contained in the magnetic base body **10** is not limited to those described above. The first metal magnetic particles **11** may contain, for example, 85 wt % or more Fe. This provides the magnetic base body **10** with an excellent magnetic permeability. The composition of the second metal magnetic particles **12** is either the same as or different from that of the first metal magnetic particles **11**. When the magnetic base body **10** contains the plurality of third metal magnetic particles (not shown), the composition of the third metal magnetic particles is either the same as or different from that of the first metal magnetic particles **11**, as with the second metal magnetic particles **12**.

(18) The surfaces of the metal magnetic particles may be coated with insulating films (not shown). The insulating films are formed of, for example, a glass, a resin, or other materials having a high insulating property. For example, the insulating films are formed on the surfaces of the first metal magnetic particles **11** by mixing the first metal magnetic particles **11** with powder of a glass material in a friction mixer (not shown). The insulating films formed of the glass material are adhered to the surfaces of the first metal magnetic particles **11** by the compression friction action in the friction mixer. The glass material may contain ZnO and P.sub.2O.sub.5. The insulating films may be formed of various glass materials. The insulating films **14** may be formed of alumina powder, zirconia powder, or any other oxide powders having a high insulating property, in place of or in addition to the glass powder. The thickness of the insulating films is, for example, 100 nm or smaller.

(19) The second metal magnetic particles **12** may be coated with different insulating films than the first metal magnetic particles **11**. The insulating films may be oxide films formed by oxidation of

the second metal magnetic particles **12**. The thickness of the insulating films is, for example, 20 nm or smaller. The insulating films may be oxide films formed on the surfaces of the second metal magnetic particles **12** by heat-treating the second metal magnetic particles **12** in the atmosphere. The insulating films may be oxide films containing oxides of Fe and other elements contained in the second metal magnetic particles **12**. These insulating films may be iron phosphate films formed on the surfaces of the second metal magnetic particles **12** by placing the second metal magnetic particles **12** into phosphoric acid and stirring. The insulating films of the first metal magnetic particles **11** may be oxide films formed by oxidation of the first metal magnetic particles **11**, whereas the insulating films of the second metal magnetic particles **12** may be coating films formed by a method other than oxidation of the second metal magnetic particles **12**.

(20) The binder **13** is, for example, a thermosetting resin having a high insulating property. Examples of the binder **13** include an epoxy resin, a polyimide resin, a polystyrene (PS) resin, a high-density polyethylene (HDPE) resin, a polyoxymethylene (POM) resin, a polycarbonate (PC) resin, a polyvinylidene fluoride (PVDF) resin, a phenolic resin, a polytetrafluoroethylene (PTFE) resin, or a polybenzoxazole (PB **0**) resin. The binder **13** may also be glass or other materials and may contain an insulating filler.

(21) The conductor **25** is formed in a pattern. In the embodiment shown, the conductor **25** is wound around the coil axis Ax (see FIG. 1). When seen from above, the conductor **25** has, for example, a spiral shape, a meander shape, a linear shape or a combined shape of these.

(22) The conductor **25** is formed by plating with Cu, Ag, or other conductive materials. The entire surface of the conductor **25** other than an end surface **25a2** and an end surface **25b2** may be coated with an insulating film. As shown, when the conductor **25** is wound around the coil axis Ax for a plurality of turns, each of the turns of the conductor **25** may be separated from adjacent turns. In this arrangement, the base body **10** mediates between the adjacent turns.

(23) The conductor **25** includes a lead-out conductor **25a1** at one end portion thereof (the end portion electrically connected with the external electrode **21**) and a lead-out conductor **25b1** at the other end portion thereof (the end portion electrically connected with the external electrode **22**). The lead-out conductor **25a1** has the end surface **25a2** at an end portion thereof, and the lead-out conductor **25b1** has the end surface **25b2** at an end portion thereof. The conductor **25** is electrically connected to the external electrode **21** via the lead-out conductor **25a1** forming one end portion thereof, and the conductor **25** is also electrically connected to the external electrode **22** via the lead-out conductor **25b1** forming the other end portion thereof.

(24) In one embodiment of the present invention, the external electrode **21** extends on a part of the first principal surface **10a**, the second principal surface **10b**, the second end surface **10c**, the first side surface **10e**, and the second side surface **10f** of the base body **10**. The external electrode **22** extends on a part of the first principal surface **10a**, the second principal surface **10b**, the second end surface **10d**, the first side surface **10e**, and the second side surface **10f** of the base body **10**. The external electrodes **21**, **22** are spaced apart from each other. Shapes and arrangements of the external electrodes **21**, **22** are not limited to those in the example shown. Both the lead-out conductor **25a1** and the lead-out conductor **25b1** lead to the first principal surface (the mounting surface) **10a** of the base body **10**, and the end surface **25a2** of the lead-out conductor **25a1** and the end surface **25b2** of the lead-out conductor **25b1** are exposed from the base body **10** through the first principal surface **10a**. That is, the end surface **25a2** of the lead-out conductor **25a1** and the end surface **25b2** of the lead-out conductor **25b1** are exposed from the base body **10** through the same surface. It is also possible that the end surface **25a2** of the lead-out conductor **25a1** and the end surface **25b2** of the lead-out conductor **25b1** are exposed from the base body **10** through different surfaces.

(25) Each of the external electrodes **21**, **22** is either entirely made of a metal or partially made of a non-metal material such as a resin. An example of an external electrode partially made of a non-metal material such as a resin is a conductive resin film. In the embodiment shown, the external

electrodes **21**, **22** are conductive resin films containing metal fillers F and a conductive resin R (see FIG. 4). The metal fillers F may include first metal fillers F1 having a relatively smaller particle size and second metal fillers F2 having a relatively larger particle size. The average particle size of the first metal fillers F1 is about 0.1 μm to 2.0 μm , and the average particle size of the second metal fillers F2 is about 2.0 μm to 15 μm . The first metal fillers F1 and the second metal fillers F2 are formed of Ag, for example. The metal fillers F may also be single size fillers. The conductive resin film may have, for example, a plating layer provided on the surface thereof. The plating layer may be composed of, for example, a single plating layer such as a Ni plating layer and a Sn plating layer, or two plating layers including a Ni plating layer and a Sn plating layer formed on the Ni plating layer.

(26) FIG. 3 is an enlarged sectional view showing, on an enlarged scale, a sectional surface around the joint between one end portion of the conductor **25** and the external electrode **21** in the coil component **1** shown in FIG. 1. As shown in FIG. 3, the coil component **1** includes a first film **23** interposed between the external electrode **21** and one end portion of the conductor **25** (the lead-out conductor **25a1**). In other words, the external electrode **21** and one end portion of the conductor **25** are electrically connected to each other via the first film **23**. The coil component **1** further includes a second film (not shown) interposed between the external electrode **22** and the other end portion of the conductor **25** (the lead-out conductor **25b1**). In the embodiment shown, the first film **23** and the second film have the same functionality and shape, and these films are formed of the same material. The following description on the first film **23** also applies to the second film unless in specific cases. In addition, FIGS. 3 and 4 for description of the first film **23** also apply to the second film.

(27) The first film **23** is interposed between the end portion of the conductor **25** (the lead-out conductor **25a1**) and the external electrode **21**, and thus the end portion of the conductor **25** and the external electrode **21** are electrically connected to each other via the first film **23**. The first film **23** electrically connects between the end portion of the conductor **25** and the external electrode **21**, but it does not necessarily physically contact with the end portion of the conductor **25** and the external electrode **21**. In addition to the first film **23**, another layer or film formed of a material having a high conductivity may be present between the end portion of the conductor **25** and the external electrode **21**. The first film **23** is formed of a material having a much lower diffusion velocity than the material forming the external electrode **21**. In the embodiment shown, the first film **23** is an oxide film. Specifically, the first film **23** is an oxide film of a metal material having the same or a lower ionization tendency than the material forming the conductor **25**. That is, the first film **23** is a metal oxide film. By way of an example, supposing that the conductor **25** is formed of copper, the material forming the first film **23** may be a copper oxide such as CuO or Cu₂O. The material forming the first film **23** is not limited to oxides of metal materials but may be, for example, oxides of Si, oxides of valve metals such as Fe, Ag, Sn, Cr, Zn, Al, Ti, Ta, Nb, Zr, Hf, Ru, In, Ga, and Ge, or oxides of alloys including these valve metals. The first metal **23** is not limited to the oxide film but may also be a nitride film, a carbonized film, or an oxynitride film. Examples of the nitride film include TiN, TaN, FeN, SiN, AlN, and so forth. Examples of the carbonized film include SiC, FeC, WC, diamond-like carbon, diamond, and so forth. Examples of the oxynitride film include TaON. The first film **23** and the second film are not necessarily formed of a single material but may be formed of a plurality of materials. In general, the diffusion velocities of the oxide, the nitride film, the carbonized film, and the oxynitride film of the above metal materials forming the first film **23** are an order of magnitude lower than that of the metal material contained in the external electrode **21**. This significantly inhibits the migration of the impurity atoms.

(28) At least a part of the first film **23** and at least a part of one end portion of the conductor **25** (the end surface **25a2** of the lead-out conductor **25a1**) are connected with each other by metallic bond. The phrase “at least a part of one end portion of the conductor **25**” mentioned here refers to some region of the end surface **25a2**. For example, the first film **23** and the end portion **25a1** may be

connected with each other by ionic bond at a peripheral portion PP of the end surface **25a2** (see FIG. 3). FIG. 3 shows an example in which the first film **23** and the end portion **25a1** of the conductor **25** are connected with each other by ionic bond at the entirety of the end surface **25a2**. In the example shown in FIG. 3, the region of the end surface **25a2** at which the first film **23** and the end portion **25a1** are ion-bonded to each other includes the peripheral portion PP.

(29) As shown in FIG. 4, a plurality of indentations are formed in the interface between the first film **23** and the external electrode **21**, and the indentations in the interface between the first film **23** and the external electrode **21** are larger than the indentations in the surface of the external electrode **21**. The indentations in the interface between the first film **23** and the external electrode **21** are formed by, for example, oxidation reaction at the surface of the first film **23**.

(30) The thickness T of the first film **23** is 10 nm to 200 nm. The thickness T of the first film **23** herein refers to the dimension of the first film **23** in the direction perpendicular to the boundary surface between the first film **23** and the end portion **25a1** of the conductor **25**. When the thickness T of the first film **23** is 200 nm or smaller, a Schottky current can pass through the first film **23**, making it possible to maintain the electrical connection between the external electrode **21** and one end portion of the conductor **25**. The thickness T of the first film **23** is preferably 50 nm or smaller. When the thickness T of the first film **23** is 50 nm or smaller, a tunnel current can pass through the first film **23**, making it possible to further reduce the electrical resistance between the external electrode **21** and one end portion of the conductor **25**.

(31) Next, a description is given of a manufacturing method of the coil component **1** as an electronic component. First, the conductor **25** formed of a metal material or the like and having a coil shape is placed into a mold, along with a mixed resin composition prepared by mixing and kneading particles including the first metal magnetic particles **11** and the second metal magnetic particles **12** with the binder **13** composed of a resin or the like. This is then compression molded such that the end surface **25a2** of the lead-out conductor **25a1** and the end surface **25b2** of the lead-out conductor **25b1** of the conductor **25** are exposed through the surface. The coil shape of the conductor **25** is not particularly limited. For example, the conductor **25** is made of a wire wound in a spiral shape, or it may be made of a planar coil instead of the wound wire. The conductor **25** may have an insulating coat. The resin in the molded product is cured to obtain the magnetic base body **10** having the conductor **25** embedded therein. The coil component may be a winding coil. In this case, for example, the base body **10** is formed by placing into a mold a mixed resin composition prepared by mixing and kneading particles including the first metal magnetic particles **11** and the second metal magnetic particles **12** with the binder **13** composed of a resin or the like, and the conductor **25** is wound around the base body **10** such that the end surface thereof is positioned on the surface of the base body **10**.

(32) Next, the surface of the magnetic base body **10** through which the end surface **25a2** of the lead-out conductor **25a1** and the end surface **25b2** of the lead-out conductor **25b1** of the conductor **25** are exposed is smoothed, and a native oxide film is removed (the first step). For example, the surface of the magnetic base body **10** is smoothed by polishing using an abrasive. The particle size of the abrasive should preferably be smaller than that of the first metal magnetic particles **11**. For example, when the average particle size of the first metal magnetic particles **11** is 30 μm , an abrasive having a particle size of 25 μm is selected. The native oxide film is removed by, for example, plasma etching using argon gas or sputtering. Any method of removing the native oxide film is available as long as it can remove oxides from the surface of the magnetic base body **10**.

(33) The next step is to form the first film **23**, contacting with one end portion of the conductor **25**, and the second film, contacting with the other end portion of the conductor **25** (the second step). In the second step, the first film **23** and the second film are formed of a material having a lower diffusion velocity than the material forming the external electrodes **21**, **22**. In the following description of the manufacturing method, it is supposed that the first film **23** and the second film are metal oxide films.

(34) To form metal oxide films having a thickness T of about 10 nm to 50 nm as the first film **23** and the second film, heat treatment is performed for 10 to 60 minutes under the condition of a temperature of 120 to 200° C. in the atmosphere. Another way to form the metal oxide films having the thickness T of about 10 nm to 50 nm is to immerse the exposed end surface **25a2** and end surface **25b2** of the conductor **25** into pure water managed at pH 7 to 13, and then dry the end surfaces in the atmosphere. Still another way to form the metal oxide films having the thickness T of about 10 nm to 50 nm is to expose the end surface **25a2** and end surface **25b2** of the conductor **25** to an atmosphere at 80% or higher RH for 10 to 30 minutes, and then dry the end surfaces. To form metal oxide films having a thickness T of about 50 nm to 100 nm as the first film **23** and the second film, the exposed end surface **25a2** and end surface **25b2** of the conductor **25** are immersed in an aqueous solution managed to be alkaline at pH above 13 and then dried in the atmosphere. The first film **23** and the second film may also be formed by sputter deposition, ion beam deposition, and vapor deposition such as CVD (chemical vapor deposition).

(35) The first film **23** and the second film may also be formed by applying, to the exposed end surface **25a2** and end surface **25b2** of the conductor **25**, a resin containing a compound that serves to form a passive oxide film. Examples of such a compound include carboxylic acids, sulfonic acids, acid anhydrides, alcohols, amines, imidazoles, phosphoniums, silanes, and so forth. By way of an example, the first film **23** and the second film may be formed by applying a conductive resin paste containing any one of a metal filler and the above compound to the exposed end surface **25a2** and end surface **25b2** of the conductor **25**, and performing heat treatment at 100 to 200° C. for a plurality of times.

(36) The final step is to form the external electrode **21** contacting with the first film **23** and the external electrode **22** contacting with the second film (the third step). The coil component **1** is manufactured through the steps described above. The coil component **1** manufactured is mounted on the circuit board by soldering the external electrodes **21**, **22** to the corresponding land portions of the circuit board.

(37) As described above, the coil component **1** includes the first film **23** and the second film. The first film is positioned between one end portion (the lead-out conductor **25a1**) of the conductor **25** and the external electrode **21**, and the second film is positioned between the other end portion (the lead-out conductor **25b1**) of the conductor **25** and the external electrode **22**. The diffusion velocity in the first film **23** and the second film is lower than that in the external electrode **21** and the external electrode **22**.

(38) In general, coil components experience thermal expansion and thermal contraction due to Joule heat produced by electric current flowing through the coil components and the ambient temperature around the coil components being used. The thermal expansion and the thermal contraction cause strain to accumulate in the coil components, possibly resulting in a fault in the coil components. The coil components are particularly required to have stable bonding between the conductor and the external electrodes. Therefore, for conventional coil components, it is supposed that oxides remaining between the end portions of the conductor and the external electrodes should preferably be removed. By contrast, the present invention addresses the reduction in the joint strength between the conductor **25** and the external electrodes **21**, **22** caused by alloying of the impurity atoms with the material forming the conductor **25**. The coil component **1** positively includes the first film **23** and the second film having a lower diffusion velocity than the external electrode **21** and the external electrode **22**. In this way, the presence of the first film **23** and the second film having a lower diffusion velocity than the external electrode **21** and the external electrode **22** inhibits movement of the impurity atoms. Accordingly, the migration of impurity atoms between the conductor **25** and the external electrode **21** and between the conductor **25** and the external electrode **22** can be inhibited. As a result, it is possible to inhibit the reduction in the joint strength between the conductor **25** and the external electrodes **21**, **22** caused by alloying of the impurity atoms with the material forming the conductor **25**.

(39) In the coil component **1** according to one embodiment of the present invention, the thickness **T** of the first film **23** and the thickness of the second film are 10 nm to 200 nm. These thicknesses allow a Schottky current or a tunnel current to flow through the first film **23** and the second film. Thus, there is less impact of the first film **23** on electric connection between the conductor **25** and the external electrode **21** (an example of such an impact is an increased resistance between the conductor **25** and the external electrode **21** produced by the first film **23**). Similarly, there is less impact of the second film on electric connection between the conductor **25** and the external electrode **22**. Accordingly, it is possible to inhibit the migration of the impurity atoms while maintaining the electric connection between the conductor **25** and the external electrodes **21**, **22**.

(40) In the coil component **1** according to one embodiment of the present invention, the first film **23** and the second film are an oxide of a metal material having the same or a lower ionization tendency than the material forming the conductor **25**. In this configuration, the first film **23** and the second film can further inhibit the migration of the impurity atoms.

(41) In the coil component **1** according to one embodiment of the present invention, indentations in the interface between the first film **23** and the external electrode **21** and the interface between the second film and the external electrode **22** are larger than the indentations in the surfaces of the external electrode **21** and the external electrode **22**. In this configuration, the adhesion between the first film **23** and the external electrode **21** and the adhesion between the second film and the external electrode **22** can be increased by the anchor effect.

(42) The method of manufacturing the coil component **1** according to one embodiment of the present invention includes the second step of forming the first film **23** and the second film such that the first film **23** is electrically connected with one end portion (the lead-out conductor **25a1**) of the conductor **25** and the second film is electrically connected with the other end portion (the lead-out conductor **25b1**) of the conductor **25**. In the second step, a material having a lower diffusion velocity than the material forming the external electrode **21** and the external electrode **22** are used to form the first film **23** and the second film. In this way, since the first film **23** and the second film are formed of a material having a lower diffusion velocity than the material forming the external electrode **21** and the external electrode **22**, movement of the impurity atoms is inhibited by the first film **23** and the second film. Accordingly, the migration of impurity atoms between the conductor **25** and the external electrode **21** and between the conductor **25** and the external electrode **22** can be inhibited.

(43) Next, a description is given of a coil component **100** according to another embodiment of the present invention with reference to FIG. 5. FIG. 8 is a perspective view schematically showing the coil component **100**. As shown, similarly to the coil component **1**, the coil component **100** includes a coil conductor **25** provided in the base body **10**, an external electrode **21** disposed on a surface of the base body **10**, and an external electrode **22** disposed on the surface of the base body **10** at a position spaced apart from the external electrode **21**. The coil component **100** is different from the coil component **1** in that it includes an insulating plate **50** and two conductors **25**. The insulating plate **50** is provided in the base body **10**, and the two conductors **25** are provided on the top-side surface and the bottom-side surface of the insulating plate **50**, respectively.

(44) Similarly to the coil component **1**, the coil component **100** includes the first film **23** and the second film. The first film is positioned between one end portion (the lead-out conductor **25a1**) of the conductor **25** and the external electrode **21**, and the second film is positioned between the other end portion (the lead-out conductor **25b1**) of the conductor **25** and the external electrode **22**. The diffusion velocity in the first film **23** and the second film is lower than that in the external electrode **21** and the external electrode **22**. Accordingly, for the same reason as with the coil component **1**, the migration of impurity atoms between the conductor **25** and the external electrode **21** and between the conductor **25** and the external electrode **22** can be inhibited.

(45) The dimensions, materials, and arrangements of the constituent elements described for the above various embodiments are not limited to those explicitly described for the embodiments, and

these constituent elements can be modified to have any dimensions, materials, and arrangements within the scope of the present invention. Furthermore, constituent elements not explicitly described herein can also be added to the above-described embodiments, and it is also possible to omit some of the constituent elements described for the embodiments.

(46) The electronic component according to the present invention is not limited to a coil component but may be a capacitor, for example. FIG. 6 is a sectional view schematically showing a capacitor 200 as an electronic component according to still another embodiment of the present invention. As shown in FIG. 6, the capacitor 200 includes a base body 210, a conductor 225 provided in the base body 210, and external electrodes 202, 203 disposed on the periphery of the base body 210. In the embodiment shown in FIG. 6, the capacitor 200 is what is called a MLCC (multilayer ceramic capacitor), and the conductor 225 includes a plurality of first electrode layers 221 and a plurality of second electrode layers 222. The first electrode layers 221 and the second electrode layers 222 alternate, and the base material 210 mediates between these layers. The portions of the base material 210 mediating between the first electrode layers 221 and the second electrode layers 222 serve as a dielectric. Similarly to the coil component 1, the capacitor 200 includes the first film 23 and the second film. The first film 23 is positioned between the external electrode 202 and the conductor 225, and the second film is positioned between the external electrode 203 and the conductor 225.

(47) Similarly to the coil component 1, the capacitor 200 includes the first film 23 and the second film. The first film is electrically connected with the external electrode 202 and positioned between one end portion of the conductor 225 and the external electrode 202, and the second film is electrically connected with the external electrode 203 and positioned between the other end portion of the conductor 225 and the external electrode 203. The diffusion velocity in the first film 23 and the second film is lower than that in the external electrode 202 and the external electrode 203. Accordingly, for the same reason as with the coil component 1, the migration of impurity atoms between the conductor 225 and the external electrode 202 and between the conductor 225 and the external electrode 203 can be inhibited.

Claims

1. An electronic component, comprising: a base body; a conductor provided inside or outside the base body; a first external electrode and a second external electrode both electrically connected to the conductor; and a first film positioned between the first external electrode and one end portion of the conductor electrically connected to the first external electrode, and a second film positioned between the second external electrode and the other end portion of the conductor electrically connected to the second external electrode, wherein a diffusion velocity in the first film and the second film is lower than that in the first external electrode and the second external electrode, and wherein the first film and the second film are nitride films, carbonized films, or oxynitride films, and wherein the nitride films include TaN, FeN, SiN or AlN, and the oxynitride films include TaON.
2. The electronic component of claim 1, wherein a thickness of the first film and a thickness of the second film are 10 nm to 200 nm.
3. The electronic component of claim 1, wherein the first film and the second film are amorphous.
4. The electronic component of claim 1, wherein indentations in an interface between the first film and the first external electrode and an interface between the second film and the second external electrode are larger than indentations in surfaces of the first external electrode and the second external electrode.
5. The electronic component of claim 1, wherein the conductor includes a portion wound around a coil axis.
6. A circuit board comprising the electronic component of claim 1.

7. An electronic device comprising the circuit board of claim 6.

8. A method of manufacturing an electronic component including a base body, a conductor provided inside or outside the base body, and a first external electrode and a second external electrode both electrically connected to the conductor, the method comprising: a first step of positioning a first end portion and a second end portion of the conductor at a surface of the base body, the first end portion being configured to be electrically connected with the first external electrode, the second end portion being configured to be electrically connected with the second external electrode; a second step of forming a first film and a second film, the first film being electrically connected with the first end portion, the second film being electrically connected with the second end portion; and a third step of forming a first external electrode and a second external electrode, the first external electrode being electrically connected with the first film, the second external electrode being electrically connected with the second film, wherein in the second step, the first film and the second film are formed of a material having a lower diffusion velocity than a material forming the first external electrode and the second external electrode, the first film and the second film being nitride films, carbonized films, or oxynitride films, the nitride films including TaN, FeN, SiN or AlN, the oxynitride films including TaON.
